전산화학 (2016년 1학기) 2차 시험 해답

(총 **6문제, 110점**)

1. **(총 20점, 각 5점)**

(a) $df(x,y) = xe^{axy} + 2ye^{axy}dy$

$$M(x,y) = xe^{axy}, \qquad N(x,y) = 2ye^{axy}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = ax^2e^{axy}$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = 2ay^2e^{axy}$$
$$M_y(x,y) \neq N_x(x,y)$$

불완전미분,

(b) $df(x,y) = by\cos(bx)dx + \sin(bx)dy$

$$M(x,y) = by\cos(bx), \qquad N(x,y) = \sin(bx)$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = b\cos(bx)$$
$$N_y(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = b\cos(bx)$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = y\sin(bx) + C$

(c) $df(x,y) = (x+y)^{-1}dx + (x+y)^{-1}dy$

$$M(x,y) = \frac{1}{x+y}, \qquad N(x,y) = \frac{1}{x+y}$$
$$M_y(x,y) = \left(\frac{\partial M}{\partial y}\right)_x = -\frac{1}{(x+y)^2}$$
$$N_x(x,y) = \left(\frac{\partial N}{\partial x}\right)_y = -\frac{1}{(x+y)^2}$$
$$M_y(x,y) = N_x(x,y)$$

완전미분, $f(x,y) = \ln(x+y) + C$

(d) $df(T,V) = C_V(T)dT + \frac{RT}{V}dV$

$$M(T,V) = C_V(T), \qquad N(x,y) = \frac{RT}{V}$$
$$M_V(T,V) = \left(\frac{\partial M}{\partial V}\right)_T = 0$$
$$N_T(T,V) = \left(\frac{\partial N}{\partial T}\right)_V = \frac{R}{V}$$
$$M_V(T,V) \neq N_T(T,V)$$

불완전 미분

2. **(총 20점, 각 10점)**

(a) $f(x,y) = \frac{1}{2}x^2 + \frac{1}{4}y^4 - \frac{1}{2}y^2$ 의 임계점을 (x^*, y^*) 라 하면, 임계점에서

$$f_x(x^*, y^*) = 0 \Rightarrow (x^*) = 0 \Rightarrow x^* = 0 \quad (1)$$

$$f_y(x^*, y^*) = 0 \Rightarrow (y^*)^3 - (y^*) = 0 \Rightarrow (y^*)((y^*)+1)((y^*)-1) = 0 \Rightarrow y^* = 0, \pm 1 \quad (2)$$

그러므로 임계점은 $(0,0),\ (0,-1),\ (0,1)$ 이다.

Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x,y) = \frac{1}{2}x^2 + \frac{1}{4}y^4 - \frac{1}{2}y^2$ 에 대하여

$$\begin{aligned} f_{xx}(x,y) &= 1 \\ f_{yy}(x,y) &= 3y^2 - 1 \\ f_{xy}(x,y) &= 0 \end{aligned}$$

i. $(0,0)$ 에서

$$\begin{aligned} f_{xx}(0,0) &= 1 \\ f_{yy}(0,0) &= -1 \\ f_{xy}(0,0) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 1 > 0 \quad \text{그리고} \quad \Delta_2 = (1) \times (-1) - (0)^2 = -1 < 0$$

$\Delta_2 < 0$ 이므로 $f(x,y)$ 는 $(0,0)$ 에서 안장점을 갖는다.

ii. $(0,-1)$ 에서

$$\begin{aligned} f_{xx}(0,-1) &= 1 \\ f_{yy}(0,-1) &= 2 \\ f_{xy}(0,-1) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 1 > 0 \quad \text{그리고} \quad \Delta_2 = (1) \times (2) - (0)^2 = 2 > 0$$

$\Delta_1 > 0$ 이고, $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(0,-1)$ 에서 극솟값을 갖는다.

iii. $(0,1)$ 에서

$$\begin{aligned} f_{xx}(0,1) &= 1 \\ f_{yy}(0,1) &= 2 \\ f_{xy}(0,1) &= 0 \end{aligned}$$

그러므로

$$\Delta_1 = 1 > 0 \quad \text{그리고} \quad \Delta_2 = (1)\times(2) - (0)^2 = 2 > 0$$

$\Delta_1 > 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $(0,1)$ 에서 극솟값을 갖는다.

3. **(총 20점)**

(a) **(5점)** $\int_{-2}^{2} \frac{x^3}{1+x^2}dx$ 는 기함수이므로 -2 에서 2 까지 적분은 0

$$\int_{-2}^{2} \frac{x^3}{1+x^2}dx = \int \frac{x^2 x dx}{1+x^2} = 0$$

(b) **(5점)** $t = x^2$ 로 치환하면, $dt = 2xdx$ 이므로

$$\int_0^{\pi/2} x\sin x^2 dx = \frac{1}{2}\int_0^{\pi^2/4} \sin t dt = \frac{1}{2}\Big[-\cos t\Big]_0^{\pi^2/4} = \frac{1}{2}\big(1-\cos(\pi^2/4)\big)$$

(c) **(10점)**

$$\begin{aligned} \int_0^b dy \int_0^a x(a-x)y(b-y)dx &= \int_0^b y(b-y)dy \int_0^a x(a-x)dx \\ &= \int_0^b (by-y^2)dy \int_0^a (ax-x^2)dx \\ &= \left[\frac{b}{2}y^2 - \frac{1}{3}y^3\right]_0^b \left[\frac{a}{2}x^2 - \frac{1}{3}x^3\right]_0^a \\ &= \frac{b^3}{6}\times\frac{a^3}{6} = \frac{a^3b^3}{36} \end{aligned}$$

4. **(10점)** $f(x,y) = xy\sin 5x + x^2\sqrt{y}\ln y + 3e^{-2x^2}$

$$\begin{aligned} df(x,y) &= \left(\frac{\partial f}{\partial x}\right)_y dx + \left(\frac{\partial f}{\partial y}\right)_x dy \\ &= \big(y\sin 5x + 5xy\cos 5x + 2x\sqrt{y}\ln y\big)dx + \left(x\sin 5x + \frac{x^2}{2\sqrt{y}}\ln y + \frac{x^2\sqrt{y}}{y}\right)dy \end{aligned}$$

5. **(10점)**

$$\begin{aligned}\frac{\partial f}{\partial \theta} &= \left(\frac{\partial f}{\partial x}\right)_y \left(\frac{\partial x}{\partial \theta}\right) + \left(\frac{\partial f}{\partial y}\right)_x \left(\frac{\partial y}{\partial \theta}\right) \\ &= \frac{2}{2x+3y}(-a\sin\theta) + \frac{3}{2x+3y}(a\cos\theta) \\ &= \frac{-2a\sin\theta + 3a\cos\theta}{2x+3y} \\ &= \frac{-2a\sin\theta + 3a\cos\theta}{2a\cos\theta + 3a\sin\theta} \\ &= \frac{3\cos\theta - 2\sin\theta}{3\sin\theta + 2\cos\theta}\end{aligned}$$

6. **(총 20점, 각 10점)**

(a) **(10점)**

$$\begin{aligned}&\int_C (2xydx + x^2dy), \qquad (C : (1,1) \to (1,8) \to (6,8)) \\ &= \int_1^8 (1)^2 dy + \int_1^6 2x(8)dx \\ &= \Big[y\Big]_1^8 + \Big[8x^2\Big]_1^6 = 7 + 280 = 287\end{aligned}$$

(b) **(10점)**

$$\begin{aligned}&\int_C (2xydx + x^2dy), \qquad (C : (1,1) \to (1,3) \to (4,3) \to (4,8) \to (6,8)) \\ &= \int_1^3 (1)^2 dy + \int_1^4 2x(3)dx + \int_3^8 (4)^2 dy + \int_4^6 2x(8)dx \\ &= \Big[y\Big]_1^3 + \Big[3x^2\Big]_1^4 + \Big[16y\Big]_3^8 + \Big[8x^2\Big]_4^6 = 2 + 45 + 80 + 160 = 287\end{aligned}$$

(c) **(5점)** (a)와 (b)의 결과에서 경로에 관계없이 선적분 값이 같으므로 완전미분이다.

(d) **(5점)** $M(x,y) = 2xy,\ N(x,y) = x^2$ 로 놓으면,

$$\left(\frac{\partial M}{\partial y}\right)_x = 2x, \qquad \left(\frac{\partial N}{\partial x}\right)_y = 2x \Rightarrow \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

이므로, $df(x,y) = 2xydx + x^2dy$ 는 완전미분이다.